



中国科学院大学
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Detection of mild depression based on multimodal data of electroencephalogram and eye movement tracking

基于脑电图与眼动追踪多模态数据检测轻度抑郁症

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Research background



A Silent Struggle: How Do We See the Unseen? (一场无声的挣扎：我们如何看见看不见的痛苦？)

Depression affects hundreds of millions, yet it often remains invisible.
Why is it so hard to detect accurately?





A Widespread Mental Health Crisis

According to the WHO, **depression** is a leading cause of disability worldwide, affecting an estimated **280 million** people globally (as of 2023).



Mild Depression: A Hidden Epidemic

Often undiagnosed, mild symptoms significantly impair daily functioning and productivity, acting as a precursor to more severe depressive episodes.



Limitations of Traditional Diagnostic

Such as scale assessments and clinical interviews, rely heavily on **subjective judgment** and are prone to **misdiagnosis and underdiagnosis**.



Core Concepts:

- **Eye-Tracking Technology 眼动追踪**: Non-invasive technology, reflects human cognitive and emotional processing, its features can reflect attentional biases in depressed patients. (Focuses on external cognitive behavior)
- **EEG (Electroencephalogram) 脑电图**: Non-invasive technology, records brain electrical activity, serves as a reliable neural biomarker for depression detection, with higher sampling frequency. (Focuses on internal brain neural activity)
- **Multimodal Data Fusion (EEG + Eye-Tracking) 多模态数据融合**: Combines the two non-invasive signals, complements their advantages, makes up for the one-sidedness of single-modal data, and provides more comprehensive information for depression detection.



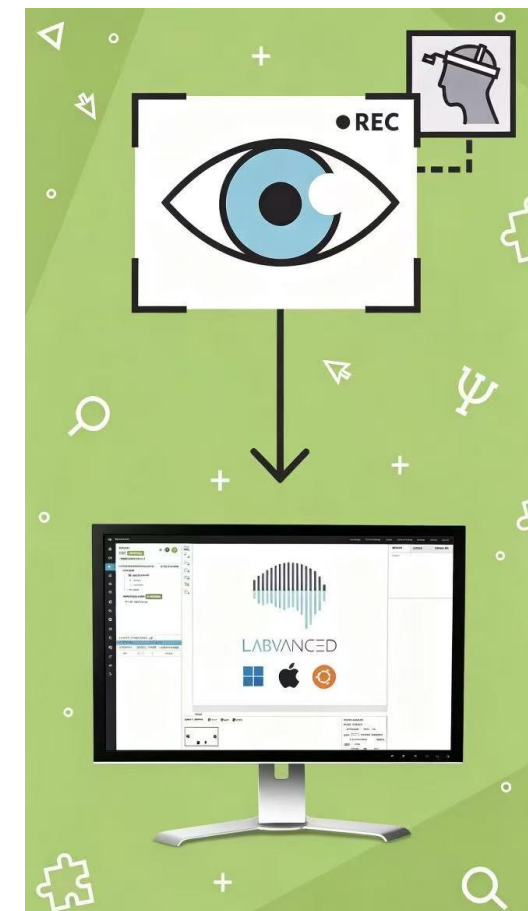
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Literature review

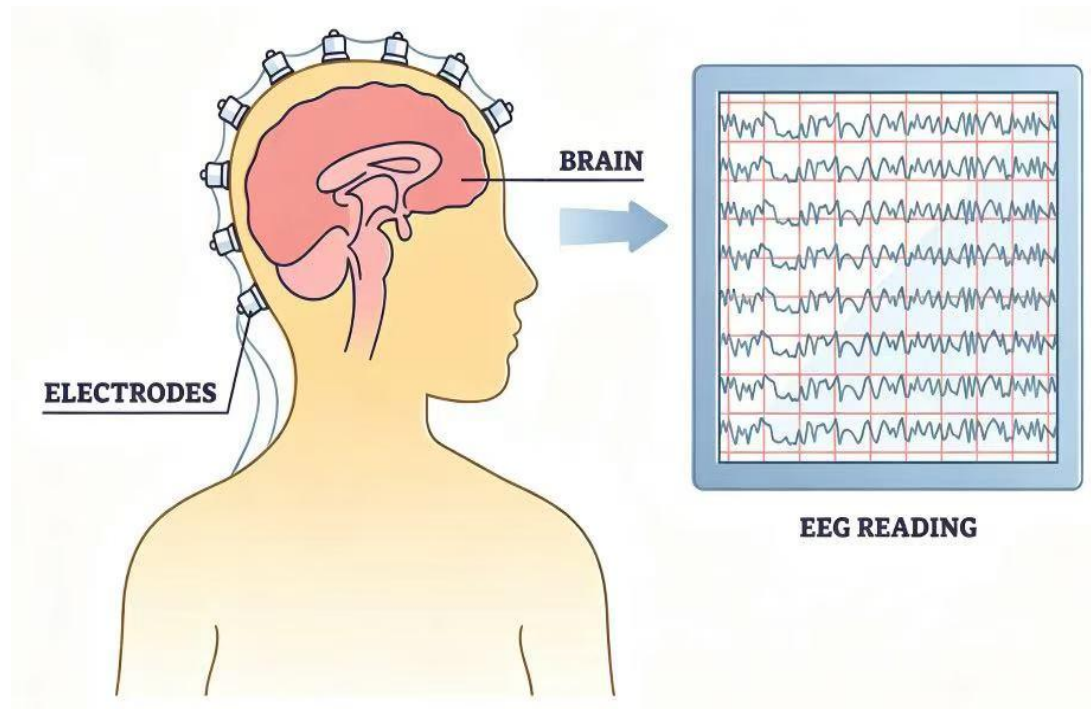


Part 1: Eye tracking in Depression Detection

Source	Methods	Results and conclusions	Limitations
Alhowinem et al. (2013)	Analyzed eye movement features	Eye metrics reflect depressive attentional biases, laid foundation for eye-tracking in depression diagnosis	Qualitative analysis only, no deep learning
Stolicyn et al. (2022)	Developed a depression assessment system	Extracted facial/eye-tracking features, 79% accuracy via SVM	No mild depression classification, limited generalization
Yang et al. (2022)	Proposed spatiotemporal clustering algorithm	Identified ROIs via eye-tracking, 76.25% accuracy (89 participants)	Small sample, low sensitivity to mild depression



Part 2: EEG in Depression Detection



- ① Mahato & Paul (2020) classified depression and healthy subjects using EEG band features and SVM, achieving 88.33% accuracy.
 - Limitation: traditional machine learning, manual feature extraction



Deep Learning

Deep learning extracts features automatically and effectively.

- ② Yin et al. (2023) proposed a Transformer-CNN hybrid model that outperforms existing methods.
- ③ Tigga & Garg (2023) introduced an attention-based Transformer model, achieving 98.67% accuracy in depression detection.



➔ With the rise of multimodal biomedical data, **Data fusion** is increasingly applied.



- ④ Ding et al. (2019) classified major depression using EEG, eye tracking and GSR data, achieving an accuracy of 79.63%.
- ⑤ Zhu et al. (2019) proposed a content-based ensemble method to fuse EEG and eye tracking data, validating its effectiveness in identifying depressed patients. The fused information is more complex and comprehensive than original single-modal data.

Limitations: While existing methods perform well, most rely on single-modal data and cannot fully capture the complex pathology of depression. Multimodal fusion remains preliminary, lacking effective strategies for EEG and eye tracking.



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Classification



**Objective auxiliary diagnosis of mild
depression in adolescents**

Primary discipline

Biomedical Engineering
Biomedical Signal Processing

Secondary discipline

Multimodal Data Fusion
Affective Computing
Brain-Computer Interface

Tertiary research direction

Intelligent Depression Detection
Deep Learning
Transformer Model
Joint Analysis of EEG and Eye Tracking



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**Issues, Gaps, Goals, logic,
Significance**



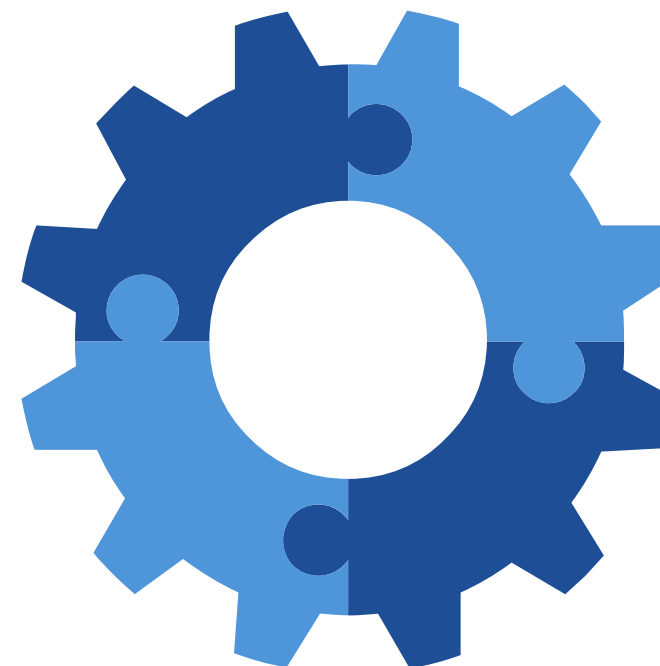
Primary Question

How can we objectively and accurately detect mild depression in adolescents using **non-invasive** biological signals?



Technical Question

What is the most effective **deep learning architecture and fusion stage** (early, intermediate, or late) to integrate heterogeneous multimodal data (EEG and Eye Tracking) for depression detection?





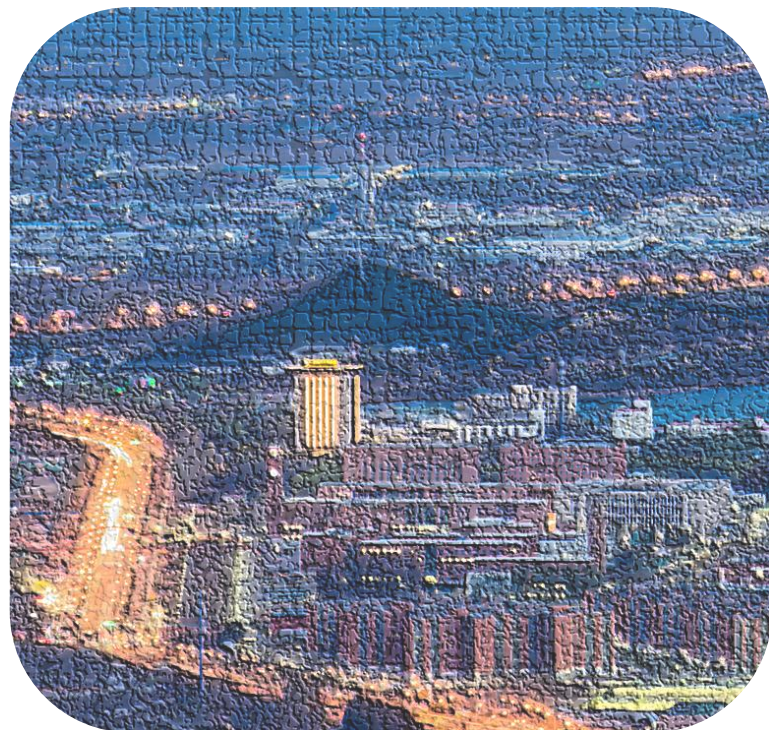
overcome the limitations of **single-mode data** in capturing physiological signal characteristics.



the **difference** between **attention maps** generated by deep learning models (e.g., Transformers) and actual **human visual attention mechanisms** has not been thoroughly investigated in the context of depression detection



there is a lack of systematic comparison and optimization of **fusion strategies** (feature-based vs. deep learning fusion) and **fusion stages** (early, intermediate, late) specifically for depression detection.



Platform Development

To develop a **multimodal signal acquisition platform** to investigate visual attention and cognitive brain activities.

Model Innovation

To propose a novel **Multimodal Transformer Network** (MTNet) to capture local and global spatio-temporal characteristics of EEG and eye-tracking data.

Strategy Evaluation

To evaluate and compare the **impact of different fusion stages** (early, intermediate, and late) on depression classification performance.

**Problem-driven** → **Method design** → **Experiment verification****I . Problem-driven**

The diagnosis of depression is highly subjective.
It is necessary to search for objective and non-invasive biological signal indicators

II . Method design

Limitations of single modality → Multimodal fusion (EEG + eye tracking)

Data acquisition & preprocessing (49 adolescents)

Method design: { Traditional ML (handcrafted features + SVM etc.)
Deep learning MTNet (Transformer + multimodal)

III . Experiment verification

49 teenagers (21 with mild depression + 28 healthy controls) participated in the driving attention paradigm (with sudden driver/vehicle collision).

Deep learning model (MTNet):

{ EEG branch
Eye tracking branch
Fusion



Clinical Value

Provides an **objective, comprehensive, and accurate** biological indicator for the **prompt detection** of adolescent mild depression.



Methodological Breakthrough

Demonstrates that the proposed **MTNet** achieves a **state-of-the-art** accuracy of 91.79%.



Fusion Stage Insight

Proves that intermediate fusion in DP is superior for processing **heterogeneous multimodal data**, effectively capturing interactions between modal features.





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References



1. Das, Abhishek, et al. "Human attention in visual question answering: Do humans and deep networks look at the same regions?." Computer Vision and Image Understanding 163 (2017): 90-100.
2. Chlasta, Karol, et al. "AI-Based Screening for Depression and Social Anxiety Through Eye Tracking: An Exploratory Study." arXiv preprint arXiv:2503.17625 (2025).
3. Zhu, Jing, et al. "An improved classification model for depression detection using EEG and eye tracking data." IEEE transactions on nanobioscience 19.3 (2020): 527-537.
4. Zheng, Zhiguo, et al. "Diagnosing and tracking depression based on eye movement in response to virtual reality." Frontiers in Psychiatry 15 (2024): 1280935.
5. Lee, S., S. Kim, and K. M. Lee. "Depression detection using virtual avatar communication and eye tracking." Frontiers in Psychiatry 13 (2022): 10699095.
6. Zhu, Jing, et al. "Multimodal mild depression recognition based on EEG-EM synchronization acquisition network." IEEE Access 7 (2019): 28196-28210.
7. Zhu, Jing, et al. "Transformer-based fusion model for mild depression recognition with EEG and pupil area signals." Medical & Biological Engineering & Computing 63.7 (2025): 2011-2027.
8. Stolicyn, A., J. D. Steele, and P. Seriès. "Prediction of depression symptoms in individual subjects with face and eye movement tracking." Psychological Medicine 52.9 (2022): 1784-1792.
9. Yang, M., et al. "Clustering based on eye tracking data for depression recognition." IEEE Transactions on Cognitive and Developmental Systems (2022).
10. Mahato, S., and S. Paul. "Classification of depression patients and normal subjects based on electroencephalogram (EEG) signal using alpha power and theta asymmetry." Journal of Medical Systems 44 (2020): 1-8.
11. Yin, F., et al. "Depression detection in speech using transformer and parallel convolutional neural networks." Electronics 12.2 (2023): 328.
12. Tigga, N. P., and S. Garg. "Efficacy of novel attention-based gated recurrent units transformer for depression detection using electroencephalogram signals." Health Information Science and Systems 11.1 (2023): 1-17.
13. Ding, X., et al. "Classifying major depression patients and healthy controls using EEG, eye tracking and galvanic skin response data." Journal of Affective Disorders 251 (2019): 156-161.
14. Zhu, J., et al. "Toward depression recognition using EEG and eye tracking: an ensemble classification model CBEM." 2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE, 2019.



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Thank you for listening!

